Chapter 6

Indexing techniques and applications

Contents

6.1. Index calculation and index arbitrage 2
6.1.1. What is an index? 2
6.1.2. Calculation method 2
6.1.3. What is index arbitrage? 4
6.1.4. Index arbitrage example 5
6.1.5. Index arbitrage in practice 5
6.1.6. Profit realization 8
6.1.7. Excursus: Exchange Traded Funds (ETFs) 8
6.1.7. Mindset of the trader 10
6.2. Leveraged exchanged traded fund 10
6.2.1. Introduction 10
6.2.2. Performance of some funds 11
6.2.3. Daily hedging demand 12
6.2.4. Special features of the Leveraged ETF 14
6.2.5. Options of leveraged ETFs 15
6.3. Recent index investing strategies 15
6.3.1. What are Smart Beta strategies? 15
6.3.2. A simple derivative strategy: Covered Call ETF 17
6.3.2. Adaptive Market Hypothesis 18
6.3.3. ESG investing 19

Author: Ka Lok Chau

Last Revised: September 2025

6.1. Index calculation and index arbitrage

6.1.1. What is an index?

Given that there are thousands of financial instruments being traded everyday, there is a need for some benchmarks in order to represent the general performance of the markets, instead of reporting the prices of each instrument. As an example, the S&P 500 index is a good indicator of the US stock market performance. Furthermore, indices can be used to judge the performance of the asset managers (c.f. section 7.1).

An index must possess three key characteristics:

- Transparent: every aspect of the index must be public information, such as the how the index is constructed, the selection criteria for the constituents and the rebalancing methodologies, pricing information and so on.
- Investable: the index should only include financial instruments such that passive investment can be carried out, i.e. the constituents of the index can be traded in the open markets.
- Systematic: index construction should be rule-based and independent of human judgment. It can be replicated exactly once the rules are known.

Traditional indices are mostly broad-based measures of stock market performance. Following the development of the financial markets and the needs to create measurable benchmarks, many kinds of specific indicators are developed which can cover a very selective sector or strategy. Some of the criteria are even based on financial theories, such as factor indices. More recently, there have been suggestions (e.g. Lo (2016) to enlarge the definition of an index, so as to include strategies which can be dynamically adjusted (such as target-date funds and other rule-based strategies).

All kinds of indices can be created and published. In Hong Kong, a major index compiler is the Hang Seng Indexes Company Limited (www.hsi.com.hk). They publish over 2200 real time and daily indices, which are organised into six categories (some examples of each category are given below):

Market: Hang Seng Index, Hang Seng China 50 Index
Factor: Hang Seng High Beta Index, Hang Seng Mid-Cap Momentum Select Index
Strategy: Hang Seng Risk Adjusted Index – HSI 20%, HSCEI Covered Call Index
Sector: Hang Seng TECH Index, Hang Seng China A Utilities Index
Sustainability: HSI ESG Index, Hang Seng Corporate Sustainability Benchmark Index
Fixed Income: Hang Seng Markit iBoxx Offshore RMB Investment Grade Bond Index

6.1.2. Calculation method

We first look at a simple example.

		Day 0		Day 1	
stock	issued shares	Closing Price	Market Capitalization	Closing Price	Market Capitalization
A	20,000	6.0	120,000	6.5	130,000
B	5,000	17.0	85,000	17.5	87,500
C	7,000	12.0	84,000	11.3	79,100
		Total	289,000		296,600

$$I_t = I_{t-1} \times \frac{\sum w_{i,t} \times S_{i,t}}{\sum w_{i,t} \times S_{i,t-1}}$$

I_t is the index level at time t, $w_{i,t}$ is the number of shares held for stock i at time t, and $S_{i,t}$ is the price of stock i at time t. If the index level at Day 0 is 100, the index level at Day 1 is: $I_1 = 100 \times \frac{296600}{289000} = 102.63$.

Basically, the index level reflects the change in the value of a portfolio which includes all the constituent stocks. However, we need to distinguish between two concepts with respect to the weighting of each stock in the index. In this example, stock A's *dollar weighting* was 120000/289000 = 41.52% as of Day 0, and was 130000 / 296600 = 43.83% on Day 1. It is evident that dollar weighting is not a constant; it changes everyday, depending on share prices.

Another related term is the *basket weight* for each stock used to replicate the index (in number of shares). This figure remains unchanged, unless there is a change in the number of shares outstanding due to technical adjustments. For example, there can be a change in the capital structure due to corporate actions such as rights issue or share split.

We can now look at a more realistic example. The table below shows a partial list of stocks required for a theoretical basket which is equivalent to 25 futures contracts of the Hang Seng Index as of Sep 22, 2025. Since share trading in Hong Kong is usually conducted in board lots, there will be rounding errors compared to the theoretically required shares. There is a tradeoff between accuracy and liquidity, as it may not be possible to complete big orders for each stock, especially for some of the relatively smaller companies within the index (c.f. the discussion in section 6.1.5.).

stock code	name	basket shares	22-Sept close	board lot	shares required for 25 futures	rounded
9988	Alibaba	15,210	159.20	100	18,925	18,900
5	HSBC	20,540	107.20	400	25,557	25,600
700	Tencent	3,412	641.00	100	4,245	4,200
1810	Xiaomi	28,682	56.10	200	35,688	35,600
939	CCB	170,809	7.45	1,000	212,530	213,000
1299	AIA Group	16,783	72.00	200	20,882	20,800
3690	Meituan	8,682	103.90	100	10,803	10,800
388	HK Exchange	1,902	442.60	100	2,367	2,400
941	China Mobile	9,809	85.40	500	12,205	12,000
1398	ICBC	130,180	5.74	1,000	161,977	162,000
1211	BYD	5,815	109.70	500	7,235	7,000
2318	Ping An	10,583	52.95	500	13,168	13,000
981	SMIC	7,102	73.35	500	8,837	9,000
9999	Netease	2,182	234.00	100	2,715	2,700
3988	BOC	112,221	4.28	1,000	139,631	140,000
883	CNOOC	24,612	19.04	1,000	30,624	31,000
9618	JD.COM	2,761	134.20	50	3,435	3,450
1024	Kuaishou	4,737	74.00	100	5,894	5,900
9992	Pop Mart	1,166	267.60	200	1,451	1,400
3968	CM Bank	6,161	46.54	500	7,666	7,500

Hang Sang Index (HSI) calculation methodology

$$I_t = I_{t-1} \times \frac{\sum w_{i,t} \times S_{i,t} \times FAF_i \times CF_i}{\sum w_{i,t} \times S_{i,t-1} \times FAF_i \times CF_i}$$

Similar to the example used earlier, the formula above indicates that the index is calculated based on the ratio of portfolio value between the current date and the previous date. Here I_{t-1} is the index level at time $t-1$, $S_{i,t}$ is the price of stock i at time t, $w_{i,t}$ is the number of shares outstanding for company i. By definition, HSI started at 100 on its base date on July 31, 1964, but there were numerous adjustments in subsequent years, such as the regular review of constituent stocks and technical adjustments due to corporate events. One minor point to note is that the index is calculated based on current weights, as terms such as $w_{i,t-1}$ is not used in the formula above.

HSI calculation makes use of two additional terms: a free float adjustment factor FAF_i to take into account of the fact that some stocks are mostly held by the majority shareholder(s) and thus their liquidity may be limited; secondly, a capitalization adjustment factor CF_i is used to limit the weight of a stock to a maximum of 8% (effective from May 2021, with a higher percentage in earlier dates; revised every quarter). The table below shows the FAF for selected stocks (as of August 20, 2025).

Code 股份代號	Stock 股份		FAF
1	CKH Holdings	長江和記實業	70%
2	CLP Holdings	中電控股	80%
3	HK & China Gas	香港中華煤氣	60%
5	HSBC	滙豐控股	100%
11	Hang Seng Bank	恒生銀行	40%
12	Henderson Land	恒基地產	30%
16	SHK Ppt	新鴻基地產	50%
27	Galaxy Ent	銀河娛樂	50%
66	MTR Corporation	港鐵公司	30%
700	Tencent	騰訊控股	70%
1299	AIA	友邦保險	100%
3690	Meituan	美團 - W	90%
9988	Alibaba	阿里巴巴 - SW	100%
9999	Netease Inc	網易	40%

6.1.3. What is index arbitrage?

A straightforward application of the arbitrage relationship is the index arbitrage strategy, which is also known as cash-and-carry arbitrage. Recall from section 1.2.2 that the fair forward value of the index is given by: $F=(I-D)\text{ x}(1+rt)$, where I=current index, D=dividends, r = interest rate, and t = time to maturity. If the futures contract is trading

above or below F, arbitrage opportunity exists. Unlike a strategy such as *statistical arbitrage* which is really not an arbitrage (discussed in section 9.2.3), index arbitrage is a real arbitrage, i.e. if the unwinding of the trades at maturity can be conducted without slippage, a guaranteed profit can be made.

The strategy can be one of two types. Three trades are needed in each operation:

- If futures is trading above F (i.e. futures trading "at a premium"), the strategy is known as *regular arbitrage*: sell futures, borrow money at rate r for a fixed term, and buy stock basket (index).
- On the other hand, if futures is trading below F (i.e. futures trading "at a discount"), the strategy is known as *reverse arbitrage*: buy futures, borrow stock basket and short sell the stocks, and lend the money received at rate r for a fixed term.

6.1.4. Index arbitrage example

The following market rates are available: spot index level = 26300, interest rate = 2.63%, expected dividend = 100 points, time to maturity = 0.0740 yrs, index futures trading at 26283 points. The fair value of futures is given by:

$$F = (26300 - 100) \text{ x } (1+rt) = 26250.99$$

Since the calculated fair value is less than the market price of the futures, the correct strategy is regular arbitrage, i.e. borrow money, buy stock basket, and sell futures at 26283. The theoretical expected profit at maturity = 26283 – 26250.99 = 32.01 points per basket (before cost).

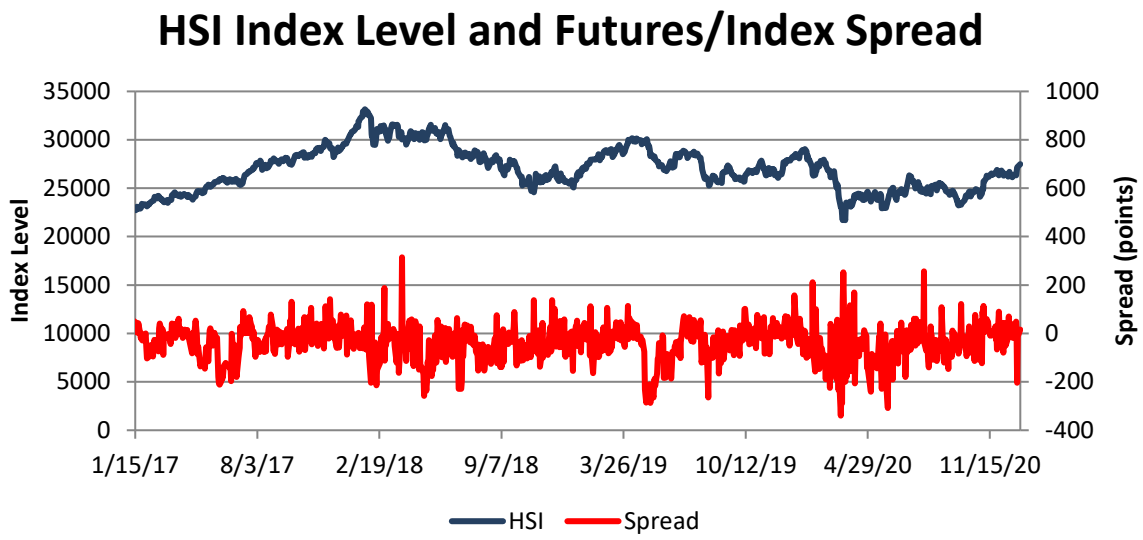
Why would arbitrage opportunity exist?

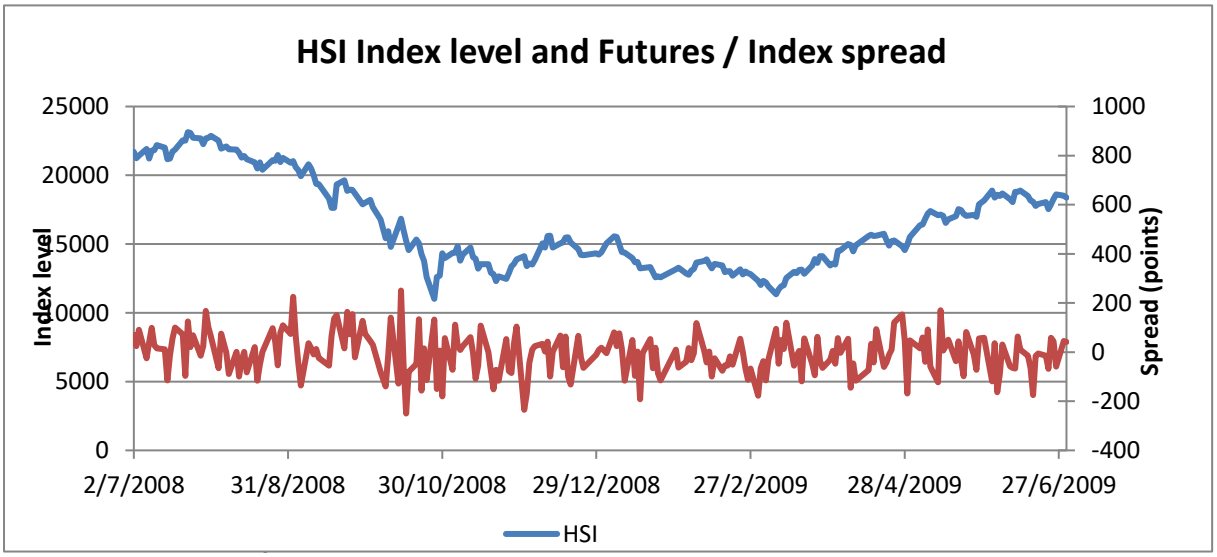
While finance theory usually assumes that arbitrage opportunities are rare, in reality they can frequently be seen, although profits can be small when various costs are taken into account. Reasons for the existence of these opportunities include:

- Convenience of futures trading as a speculative tool, and thus there can often be a temporary discrepancy between the cash and futures markets if sensitive news come out.
- Some stocks within the index may be very illiquid which would affect index calculation.
- Regulatory issues – it is not possible to short sell in certain Asian markets, therefore reverse arbitrage could not occur, and futures could trade well below its fair value. Companies which have enough inventories of stocks may be able to take advantage of this situation.
- Technical issues – limits in stock price movements exist in some markets, therefore futures could move more than the cash market when extreme moves are seen. For example, stocks may be limited to move only 5%, whereas theoretically the futures price can move more.

6.1.5. Index arbitrage in practice

In a well-behaved market, the premium and discount of the futures prices around the cash index level fluctuate from one side to another. This is also evident during the volatile periods in the most recent financial crisis in late 2008 / early 2009.



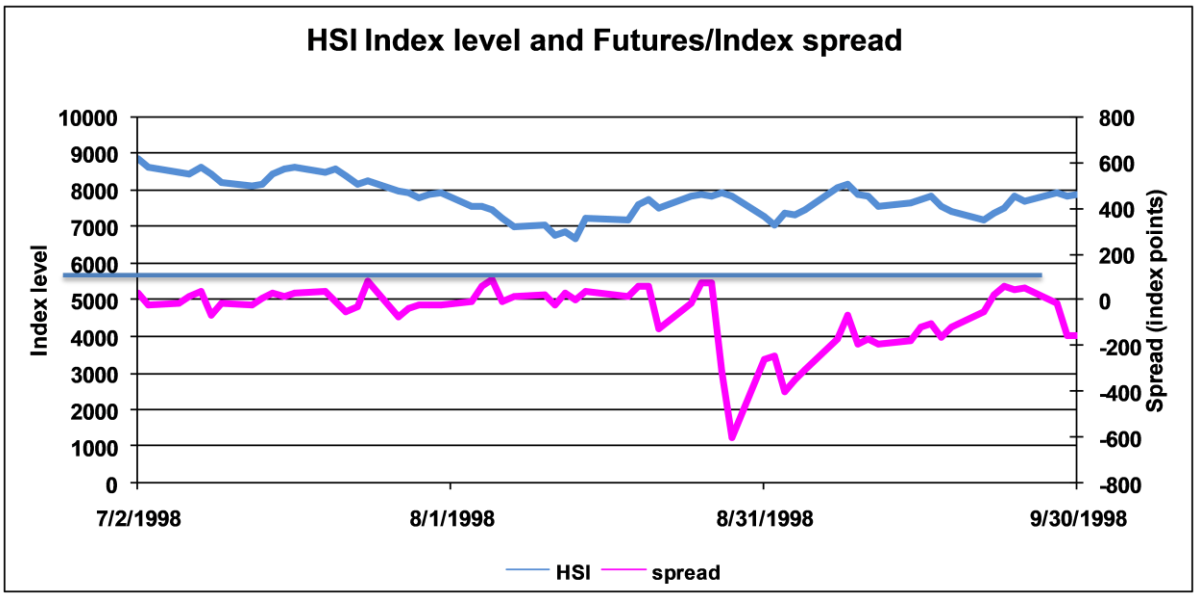


In engaging the strategy, a number of practical issues need to be taken care for:
Most indices are only updated in regular intervals. For example, Hang Seng Index is published every 2 seconds using the last traded prices of each stock, and the Shanghai Composite Index also gets updated every 2 seconds. Instead of relying on the published index, one must implement the exact formula so that real time calculation of index value is available.

In order to calculate the actual cost of the replication strategy, we need to use the correct side of the stock bid/ask price. For a higher chance of executing the trades, we can use all bid prices if we are selling stocks, and all ask prices if we are buying stocks.
Other market parameters also need to be accurate, including the spread between borrow and deposit rates, and the best estimation of future dividends (if not announced already).
Electronic trading facilitates the simultaneous execution of multiple stock orders, but one must know the market well and check the liquidity of stocks, especially for some of the minor components in the index. For example, is it possible to complete the buy/sell orders

when we need to trade stock baskets worth 50 futures contracts? Sometimes we may have to make use of a smaller "tracking" basket to represent the index (c.f. S&P 500 has 500 constituent stocks). However, this would certainly increase the slippage (i.e. tracking error). Another cause of slippage is the rounding errors due to board lots. If the trade size is too small, the resulting basket would deviate from the theoretical value because of board lot constraints.

As of August 2025, in Hong Kong the transaction fee is 0.1085% of transaction amount (roughly equal to 27 index points when HSI is at 25000). Therefore, the arbitrage profit should at least be twice this level in order for the strategy to become profitable, even assuming no further slippage in profit realization (but see the next section for a more realistic trading strategy).



A reverse arbitrage strategy can be more risky, because one needs to borrow stock and short sell, whereas the stock borrowing may not always become available. One extreme example was the incident in Hong Kong in August 1998.[1] From the diagram above, we can see that shortly before the maturity of the August contract (Aug 30, 1998), futures was trading at a discount of almost 7% to the cash index level. However, stocks could not be borrowed. Those companies that engaged the reverse arbitrage strategy earlier in the month (when the discount was at 3% to 4%, which in itself was almost a record level) were forced to close out the position before maturity, and suffered a great loss as a result.

Refer to the example in section 6.1.2, the HSI closed at 26344.14 points on Sep 22, 2025. For the equivalent of 25 futures contracts, the size of the basket is: 25 x 26344.14 x 50 = $32930175 ($50 being the value per index point). Using the proportions given in the "basket shares" column, the size of one basket is the summation of (shares outstanding x stock price), which is $26465832.63 in this example. Hence 25 futures contracts is equivalent to 32930175/26465832.63 = 1.2443 baskets. This is the multiplier used to work

[1] c.f. Paul Draper and Joseph K.W. Fung, "Discretionary Government Intervention and the Mis-pricing of Index Futures," *Journal of Futures Markets*, Special Issue 23(12):1159-89, 2003.

out the exact number of shares for each stock. For example, the theoretical number of shares of Tencent to be traded = 3412 x 1.2443 = 4245 shares. Once we obtain the figure for each stock, we can apply the board lot constraints to work out the sizes of the orders.

6.1.6. Profit realization

Once an arbitrage trade has been carried out, the theoretical arbitrage profit is "locked in." However, unless the position is closed successfully, the profit could not be realized. Given that the arbitrage condition is only satisfied at maturity, the profit can be realized only if the stocks are unwound at exactly at the price level of the futures final settlement price, because it is only at the maturity date when the futures price and the underlying index level are guaranteed to converge.

There are a few different ways to realize the profit. The straightforward way is to unwind the whole trade at maturity. Depending on the market, there may be difficulties in execution. For example, one needs to know exactly how futures settlement price is determined. In Hong Kong, the Expected Average Settlement Price (EAS) is based on calculating the index level at every five minute interval and taking the average throughout the day. Thus the correct unwind strategy is to liquidate the stock basket in equal proportions at every five minute interval. This is impossible to achieve in practice, and thus execution slippage would occur. On the other hand, in a market like Tokyo, the opening print is used in deciding the settlement price of the futures contract. Therefore it is guaranteed that the whole stock basket order can be completed other than the transaction costs to be paid.

Another way to realize the profit is to look for market opportunity to reverse the initial trades. For example, if the original trade was carried out when futures was trading at a premium, i.e. a regular arbitrage, reversing the trades means that the positions would be unwound when an opportunity for a reverse arbitrage exists. This method of unwinding is potentially the most profitable, but sometimes there may not be such an opportunity before the current futures contract expires. In this case, one could roll over to the next futures contract (maybe at a gain or loss) in order to wait for a good trading opportunity.

6.1.7. Excursus: Exchange Traded Funds (ETFs)

Similar to mutual funds, Exchange Traded Funds (ETFs) are collective investment products where an investor can buy shares in an ETF to own a proportional interest in the pooled assets. ETFs are traded in continuous markets in stock exchanges, i.e. they are traded like stocks. The liquidity of these products are often higher than the corresponding mutual funds.

Most ETFs are index funds – they are created to track the performance of published indices, which are designed to capture systematic factors over idiosyncratic factors. The management of the funds are non-discretionary and are rule-based. As a result, most ETFs benefit from lower costs and transparency in their holdings. ETFs have grown substantially in many markets, and represent a significant proportion of the daily trading volume. For example, the daily turnover of unit trusts (mostly ETFs) represent about 12.6% of the trading volume in Hong Kong Exchange in July 2025.

The largest and most successful ETF in Hong Kong is the Tracker Fund of Hong Kong (stock code 2800 and 82800). It was established on Nov 11, 1999, when the Hong Kong government needed to sell its substantial shareholdings after its intervention in the stock market during the Asian financial crisis in 1998. While the Hang Seng index was widely popular in the financial market, it was not directly investable (other than OTC products such as mutual funds). The objective of the Tracker Fund is to provide an instrument that can generate returns closely matching the corresponding performance of the Hang Seng Index.

Over the years, the Tracker Fund has become the largest and most liquid ETF trading in Hong Kong. The ongoing annual charges and fees are only 0.066% (as of August 2025), with the rolling 1-year tracking error of 0.0188%. It is also exempted from stamp duty (0.10% as of 2025), making it an attractive alternative for investments related to a basket of shares. As the Tracker Fund only pays dividends twice a year and the constituent stocks of the Hang Seng Index pay dividends throughout the year, the price of the tracker fund units represents the accumulative dividends from the last ex-dividend date. For example, as of 23/9/2025, the Hang Seng Index closed at 26159.12 points, and Tracker Fund closed at $26.80 (with a multiplier of 1000). This is not due to any substantial tracking error, as the extra value comes from the dividends accumulated from April (when the last dividend was announced and paid).

Institutional investors can make use of the Tracker Fund to create arbitrage positions. In addition to buying and selling the Tracker Fund through the open market, units can be created or redeemed directly through participating brokers, by exchanging the units with a basket of stocks. On each trading day, the composition of an index basket is published, representing the number of shares for each stock that would make up the equivalent of 1000000 units of the tracker fund (or the equivalent of 20 futures contracts). An investor can deliver multiples of 1000000 units in exchange for shares, or deliver the shares (in the correct proportion) in exchange for 1000000 units of the Tracker fund, and no stamp duty needs to be paid. The table below shows a partial list of the stock basket as at September 22, 2025:

Code 股份代號	Stock 股份		Shares
9988	Alibaba	阿里巴巴	15,210
5	HSBC	滙豐控股	20,540
700	Tencent	騰訊控股	3,412
1810	Xiaomi	小米集團	28,682
939	CCB	建設銀行	170,809
1299	AIA Group	友邦保險	16,783
3690	Meituan	美團	8,682
388	HK Exchange	香港交易所	1,902
941	China Mobile	中國移動	9,809
1398	ICBC	工商銀行	130,180
1211	BYD	比亞迪股份	5,815
2318	Ping An	中國平安	10,583
981	SMIC	中芯國際	7,102
9999	Netease	網易	2,182

6.1.7. Mindset of the trader

From the above discussion, we can see that there are many practical issues which would affect the potential profit of the strategy. In fact it is never an arbitrage in most markets. Typically a trader would open the position at a favorable spread, but may not take into account of unwind costs. Although supposedly the stocks can be lent if one is doing a regular arbitrage, normally one would not use stock borrow rate in the calculation of fair values. The closing out would be done when the market goes in an opposite direction "at some point" (e.g. futures contract changes from being trading at a premium to trading at a discount). If this opportunity is not available in the current month, the futures contract is rolled over to the next month. This roll trading of the futures contract can also be an opportunity for the trader to make money (or lose in adverse conditions).

Sometimes the index arbitrage operation is being combined with the portfolio trading desk. In that case, the reason for entering into an index arbitrage position is the ability to hold the arbitrage portfolio as a buffer for stock holding. These stock positions can then be used in other operations, e.g. for the cases where stock borrow is difficult.

For some indices, instead of a "pure" arbitrage position, a trader can choose to only enter into positions in some (but not all) shares. An optimization algorithm is required for index replication. An experienced trader can also rely on stock trading skills to enhance the arbitrage profits.

6.2. Leveraged exchanged traded fund[2]

6.2.1. Introduction

Exchange Traded Funds (ETFs) have been very popular in recent years. These instruments are seen to be transparent investment vehicles as their performances would track the underlying indices closely. Another product which appears to be similar to the vanilla ETF, known as the leveraged ETF, has been listed in the US markets since 2006.

There had been great mis-understanding about the risk and behavior of this product among many investors, including professional traders and investment advisors. Originally these products are supposed to be designed for short term traders and hedge funds only, but many investors treat these to be leveraged bets of the underlying indices. It turned out that in the US, most of these ETFs underperformed the underlying indices, and some of the funds suffered big (unexpected) losses during the financial crisis in 2008 and early 2009. In Asia, South Korea, Japan, Taiwan have listed leveraged ETFs, and some of these funds performed extremely well in 2010 when the market rebound.

Hong Kong has finally introduced a limited set of these funds in June 2016. To avoid mis-representation, these products are not allowed to use the name "ETF." As of August 2025, there are 31 leveraged ETFs listed in the Hong Kong Exchange based on overseas equity and commodity, including Nasdaq 100, a Taiwan index, Gold, Bitcoin, and some US stocks such as NVDA, TSLA, COIN. An example is CSOP NASDAQ-100 Daily (−2x) Leveraged Product (stock code 7568). Except for selected funds over short periods, turnover of some of these funds have been quite low, probably due to the limited interest in foreign indices. From Dec 23, 2016, approval has been given to list ETFs based on liquid Hong Kong indices

[2] Good references include Cheng and Madhavan (2009) and Avellaneda and Zhang (2009).

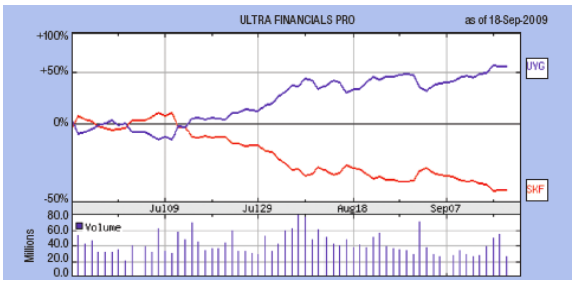
(such as HSI and HSCEI), and more recently ETFs on other indices such as A50, CSI300, Hang Seng Tech are also approved. 9 such products were listed as of August 2025, based on 2x, –1x, and –2x leverage. The biggest fund is CSOP HS TECH (stock code 7226), tracking the 2x performance of Hang Seng TECH Index, with total assets of around HKD 10.7 billion.

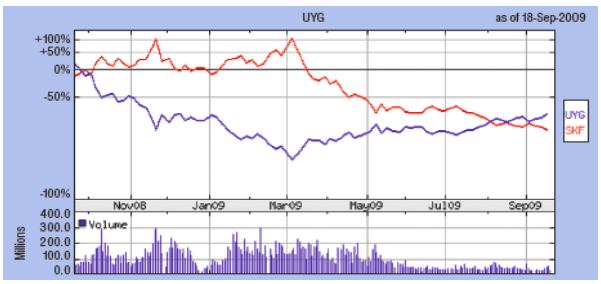
In general, leveraged ETFs are available for sectors including equities, commodities, fixed income, and foreign exchange. Basically, the daily return of the ETF is $R = x \times I$, where x is the leveraged factor and I is the daily index return. Typically x = 2 or 3 (positive leveraged) or x = –1, –2, –3 (inverse leveraged). For example,

- Proshare UltraPro Dow30 (UDOW): 3x leveraged on Dow Jones Index
- Proshare UltraPro Short Dow30 (SDOW): –3x leveraged on Dow Jones Index

In the ETF prospectuses, it is stated clearly that the return horizon is only 1-day. However, the misunderstanding is that the fund can track the long term performance of the index. An explanation is given in section 7.2.4.

6.2.2. Performance of some funds





The above diagrams show the performances of two ETFs in 2008/9.[3] These two funds track the Dow Jones Financial index offered by Proshares (UYG and SKF); one is 2x, the other is –2x. The left diagram shows the fund prices between June and September 2009. The index went up during this period, and the performances of the two funds seem to be a mirror image of each other, which is what one might expect to see. However, if we look at a longer term picture, from Sep 2008 to Sep 2009, we see that both funds lost money during the period. This may seem surprising, because if one fund returns 2x the index while the other returns –2x, how is it possible that both can lose money within one year? To be fair, this is not due to the bad management of the fund manager as the performance is not due to tracking error.

We show two more examples of extreme performances:

	Naïve return expectation	Actual return	Difference
Russell 2000 Index (RTY)	–15.8%	–15.8%	
Russel 2000 Ultra Long 2x (UWM)	–15.8% x 2 = –31.6%	–52.7%	–21.1%
Russell 2000 Short (RWM)	+15.8%	–16.5%	–32.3%
Russell 2000 Ultra Short 2x (TWM)	+15.8% x 2 = +31.6%	–51.3%	–82.9%

[3] Source: Avellaneda and Zhang (2009).

During the period between January 2008 and December 2009, the Russell 2000 index dropped 15.8%. The fund which is supposed to return 2x the performance dropped 52.7% in this period. Even the investors who hold the two funds which return –1x and –2x lost substantially.

Date	FXI	FXI return	FXP	FXP return
10/27/2008	19.29		183.01	
10/28/2008	23.15	20.0%	118.70	-35.1%
10/29/2008	22.34	-3.50%	115.48	-2.71%
10/30/2008	25.48	14.06%	87.03	-24.64%
10/31/2008	24.99	-1.92%	89.00	2.26%
11/3/2008	25.28	1.16%	86.00	-3.37%
11/4/2008	27.02	6.88%	75.20	-12.56%
11/5/2008	24.48	-9.40%	87.00	15.69%
11/6/2008	22.54	-7.92%	101.99	17.23%
11/7/2008	25.43	12.82%	76.32	-25.17%
11/10/2008	26.43	3.93%	69.50	-8.94%
11/11/2008	24.88	-5.86%	76.66	10.30%
11/12/2008	23.92	-3.86%	82.58	7.72%
11/13/2008	27.47	14.84%	60.40	-26.86%
11/14/2008	24.98	-9.06%	69.68	15.36%
11/17/2008	24.93	-0.20%	69.96	0.40%
11/18/2008	24.11	-3.29%	74.61	6.65%
11/19/2008	22.13	-8.21%	86.23	15.57%
11/20/2008	21.02	-5.02%	93.00	7.85%

The kind of behavior described here is not uncommon. In another example, FXI tracks the FTSE/Xinhua China 25 Index, FXP is –2x this index. We can see that the daily movements do not match the claimed performance exactly, but the difference after one month is surprising. While FXI total return is about 9% throughout this period (from 19.29 to 21.02), FXP's return is –49.2% (from 183.01 to 93.00).[4]

6.2.3. Daily hedging demand

It may be against intuition that the hedging that needs to be carried out would be in the same direction no matter whether it is leveraged or inverse leveraged. This is analogous to the case for options, where both in the case of short call or short put, the position is short gamma.

Assume that the NAV of a fund at time t_n is $A(t_n)$ with leverage x (which means that original exposure $L(t_n)$ is $xA(t_n)$), and the return for the period between t_n and t_{n+1} is r_n. Before re-hedging (at time t_{n+1}), the exposure is:

$$L_{BR}(t_{n+1}) = L(t_n)\,[1 + r_n] = xA(t_n)[1 + r_n]$$

whereas the NAV of the fund is

$$A(t_{n+1}) = A(t_n)[1 + xr_n]$$

which would require a new exposure of

$$L(t_{n+1}) = xA(t_{n+1}) = xA(t_n)[1 + xr_n]$$

Therefore the re-hedging that needs to be carried out

$$\begin{aligned}\Delta(t_{n+1}) &= L(t_{n+1}) - L_{BR}(t_{n+1}) \\ &= xA(t_n)[1 + xr_n] - xA(t_n)[1 + r_n] = (x^2 - x)A(t_n)r_n\end{aligned}$$

[4] Source: Euan Sinclair, *Volatility Trading*, 2nd ed. (2013), p.232.

Note that, in this expression, the term ($x^2 - x$) is always positive for any integer x if x is not equal to 1. Therefore, no matter whether the fund has positive leverage or inverse leverage, the hedging action is the same, i.e. increase exposure on a day when the index goes up, and decrease exposure on a day when the index goes down.

Case 1	$S(t_n)$	$A(t_n)$	$L(t_n)$	$L_{BR}(t_n)$	$\Delta(t_n)$
n=0	100	100	200		
n=1	90	80	160	180	−20
n=2	99	96	192	176	+16
Case 2					
n=0	100	100	-100		
n=1	90	110	-110	-90	−20
n=2	99	99	-99	-121	+22
Case 3					
n=0	100	100	-200		
n=1	90	120	-240	-180	−60
n=2	99	96	-192	-264	+72

The hedging action is shown in the examples above. The parameters are:

- Case 1: $x = 2$; Case 2: $x = -1$; Case 3: $x = -2$
- index returns on day 1 and 2 are −10% and +10%

We use case 1 as an example to illustrate how the numbers in this table are calculated. In order for it to return 2 times the performance of the index (because $x = 2$), we need the portfolio to have an exposure that is equal to 2 times the asset level, e.g. by holding a position in a futures contract with the required notional amount. That's why $L(t_n)$ would need to be maintained at 2x $A(t_n)$. At the first time step when n=0, the asset level is 100, and $L(t_0) = 200$. At time n=1, the stock index $S(t_n)$ has moved down 10% (from 100 to 90). Hence the exposure $L(t_0)$ would have the same performance, i.e. moved down 10% from 200 to 180, which is denoted as $L_{BR}(t_1)$ above (the exposure before re-hedging). This loss of 20 (=200−180) would be reflected in the change in the asset level $A(t_1)$, so that $A(t_1)$ is now 80 (=100−20).

Imagine an investor starts to hold a position in this ETF at n =1. Currently the asset level is $A(t_1)$=80. Using the logic as above, the required exposure in this fund $L(t_1)$ should be 2x $A(t_1)$ = 160. However, from the period above, we know that the exposure level (before re-hedging) is 180, so the proper action for the fund manager to take is to reduce the exposure from 180 to 160, i.e. $\Delta(t_1) = -20$.

At n=2, the index has moved up 10% (from 90 to 99). Hence $L_{BR}(t_2)$ = 160 x 110% = 176, and $A(t_2) = A(t_1) + 16 = 96$. The new required exposure $L(t_2)$ is 96x2 = 192, and the re-hedging action is thus 192−176 = +16.

Basically, the re-hedging required = $(x^2 - x)A(t_n)r_n$. At n =0, r_0 =−10% and x=2, the re-hedging required is equal to (4−2)x100x(−10%) = −20. We can see that the directions of the trades are identical in every case.

6.2.4. Special features of the Leveraged ETF

We would explain why theoretically the ETF's performance can be so different from the long term index performance. Assume we have an ETF that pays the return of the index with an initial level of 100. Furthermore, assume that the return of the index is down 10% on day 1, and up 10% on day 2. Thus the index levels on days 0 to 2 are: 100, 90, 99. We can immediately see that with two equal % moves, the ETF loses 1% over 2 days. In fact, if there is an ETF that has a leverage of +2, it would have lost 4% (100, 80, 96). This is simply because $(1-x)(1+x) = 1-x^2 < 1$ where x is the daily return. Therefore if the underlying index is volatile, an ETF tracking its daily percentage movement would show a drag in the performance even though the final index level may not move much.

Another characteristic of the leveraged ETF is that its performance is path dependent and is not related to the total return in the monitoring period. A simple two period example is shown below:

Path 1	$S(t_n)$	$A(t_n)$	$L(t_n)$	$L_{BR}(t_n)$	$\Delta(t_n)$
n=0	100	100	200		
n=1	90	80	160	180	−20
n=2	99	96	192	176	+16
Path 2					
n=0	100	100	200		
n=1	100	100	200	200	0
n=2	99	98	196	198	−2

In this example, the leverage of the fund is x = 2. In both these paths, the index level ends at 99, whereas the NAVs of the fund are different (96 and 98).

If the index follows a geometric Brownian motion, it can be shown that over a period of time t_n, the return of the fund's NAV is given by:[5]

$$\frac{A(t_n)}{A(t_0)} = C\left[\frac{S(t_n)}{S(t_0)}\right]^x, \; C = \exp\left[\frac{(x-x^2)\sigma^2 t_n}{2}\right]$$

Note that the scaling factor C is less than one as $(x-x^2) < 0$, but will never be negative, i.e. the fund's NAV will not drop below 0.

The above formula provides some clues for the long term erosion of return of the leveraged ETF. The leveraged ETF will outperform the static ETF only when volatility is low and the underlying asset (or index) moves in a constant trend. For example, when the index goes up, the re-balancing of the ETF effectively means that the exposure would be rising in a rising market, hence it will outperform the ETF which is not re-balanced. This is

[5] This result is given in both Cheng and Madhavan (2009) and Avellaneda and Zhang (2009).

seen in the term $[S(t_n)/S(t_0)]^x$ which tends to dominate (in the short run), and this is a convex relationship.

However, as shown in the examples above, the leveraged ETF cannot track index returns for periods longer than one day, and the leveraged ETF almost certainly underperforms the index / asset in the long run. Even ignoring tracking errors and transaction costs, the return depends on the index volatility and the time horizon. For example, if $x = -2$, $\sigma = 30\%$, $t_n = 5$, index return is -10%, $C = 0.259$. The fund value is only 32% of its original value even though the index has only dropped 10%.

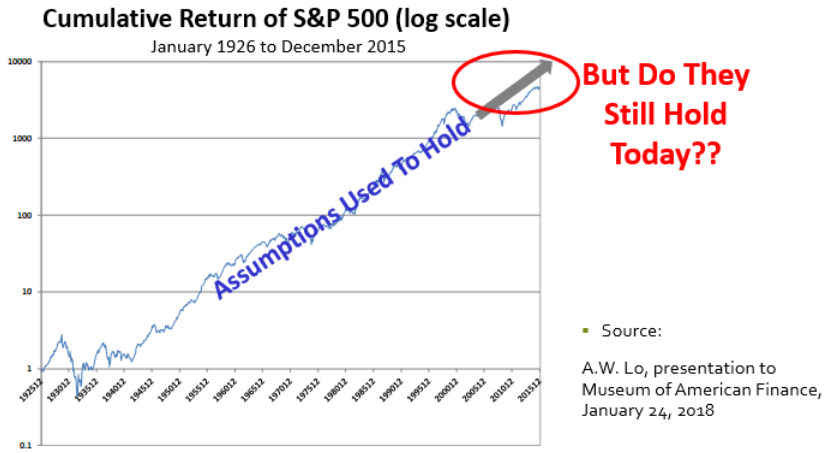
6.2.5. Options of leveraged ETFs

Given the popularity of leveraged ETFs, options on certain leveraged ETFs are also available in the market. For an option based on a leveraged ETF on the S&P 500 index, clearly the implied volatility should somehow be linked to the volatility of a non-leveraged ETF on the same index. However, empirically the connection is not very obvious, as the market does not seem to be efficient. This is an area which is still undergoing development. More information can be found in Ahn, Haugh and Jain (2014), Leung and Sircar (2014), and the detail study in Leung and Santoli (2016).[6]

6.3. Recent index investing strategies

6.3.1. What are Smart Beta strategies?

According to CAPM, beta is the only source or risk, and extra return can only be generated by taking more risk. As we have seen earlier, a rational investor should not choose investments other than index funds, because long term outperformance of active managers is not achievable.

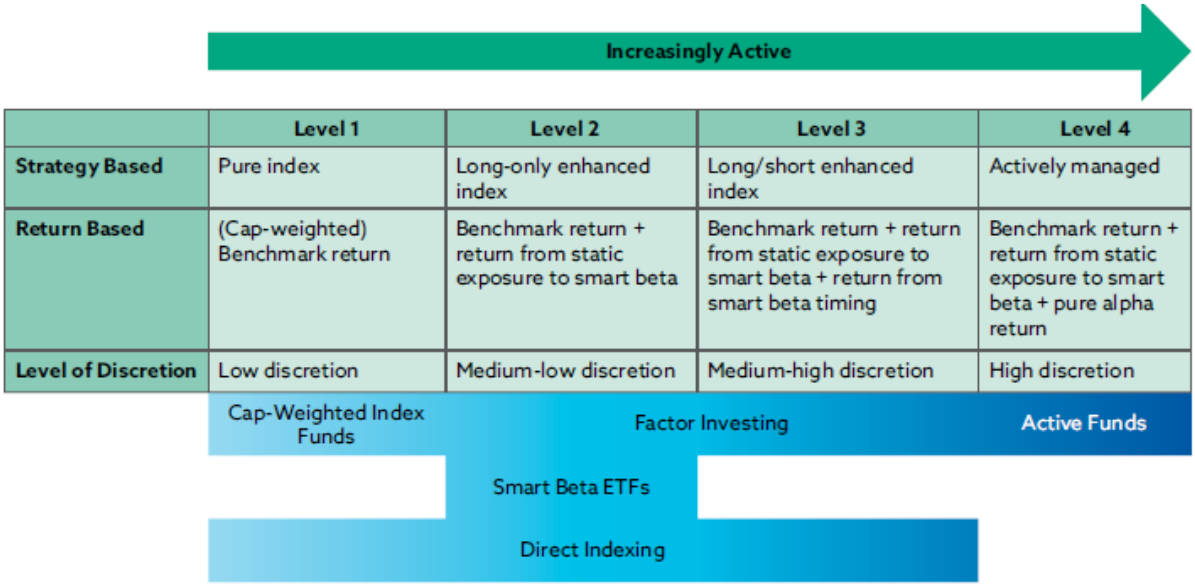


The theory has worked in the past. But should it still be valid today? In recent years, a great variety of strategies have been introduced, utilizing a diversified source of data which are not available before. These "smart beta" strategies claim to generate excess returns, with risk levels similar to investments in market index funds. On the other hand, unlike actively managed funds, these strategies are carried out based on pre-defined rules, which

[6] Tim Leung and Marco Santoli, *Leveraged Exchange-Traded Funds: Price Dynamics and Options Valuation*, Springer Finance (2016).

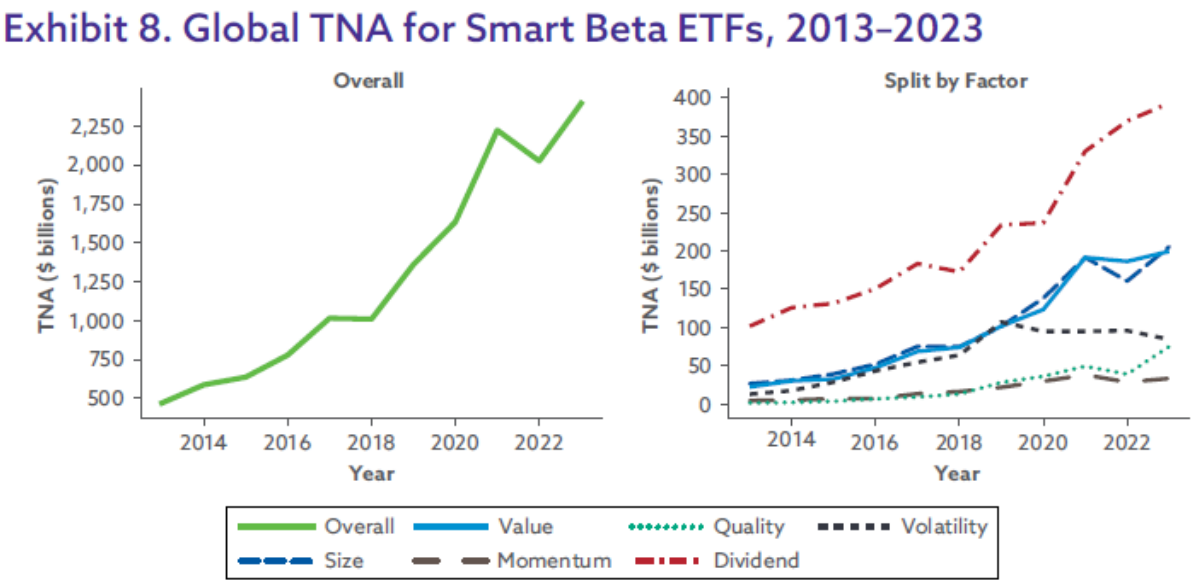
do not depend on human judgement or discretion in adjusting the trading positions. The key question is, can we identify indices with a sustainable and sufficiently attractive risk premium?

The term "smart beta" had been used in various contexts and includes a wide range of strategies. Doyle and Hayman (2024, p.30) suggest the following classification framework:



	Level 1	Level 2	Level 3	Level 4
Strategy Based	Pure index	Long-only enhanced index	Long/short enhanced index	Actively managed
Return Based	(Cap-weighted) Benchmark return	Benchmark return + return from static exposure to smart beta	Benchmark return + return from static exposure to smart beta + return from smart beta timing	Benchmark return + return from static exposure to smart beta + pure alpha return
Level of Discretion	Low discretion	Medium-low discretion	Medium-high discretion	High discretion

Following a more traditional definition, the following diagram (from Doyle and Hayman(2024)) shows the popularity of different kinds of smart beta ETFs in recent years:



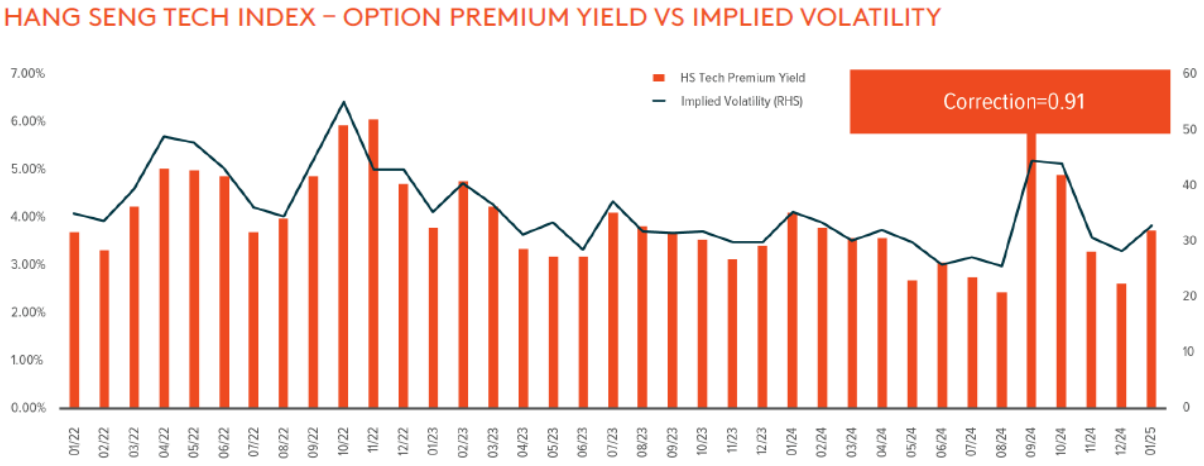
Note: Global smart beta ETFs include the universe of smart beta ETFs contained in the Lipper database.

Source: LSEG Lipper.

(TNA stands for Total Net Assets)

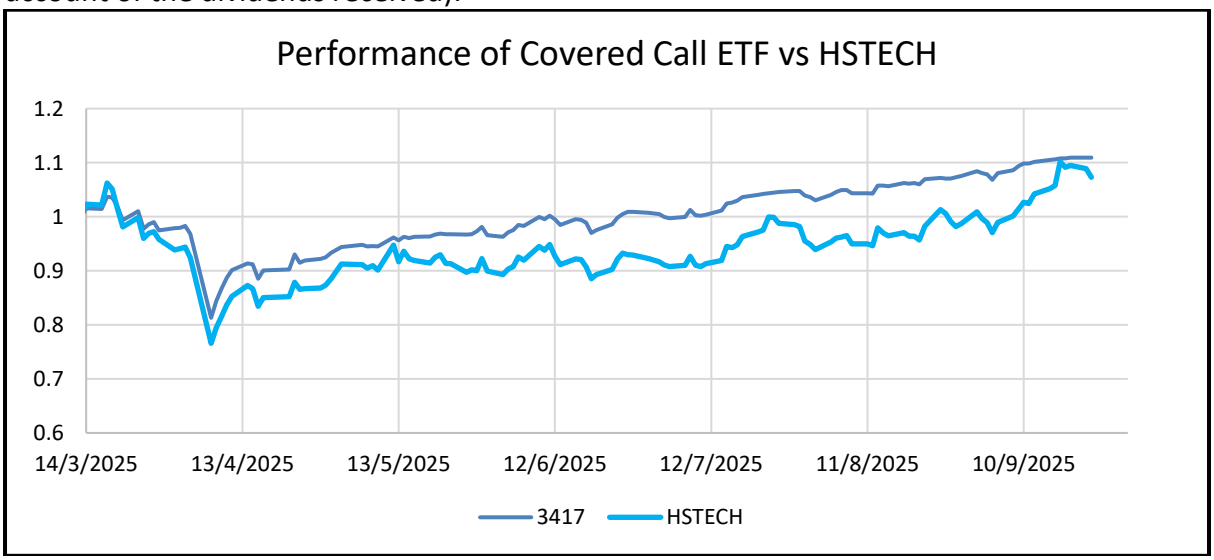
6.3.2. A simple derivative strategy: Covered Call ETF

A long-standing derivative strategy is the covered call (c.f. section 1.3.3), and the strategy is available to investors through its implementation in the form of ETF. A recent example is the Global X Hang Seng Tech Index Covered Call Active ETF (stock code 3417). This is an ETF listed in Hong Kong, based on the popular Hang Seng Tech Index (for Chinese technology stocks).



The fund will normally maintain a short position in a 1-month at-the-money call option on the index. After the option expires, a new position will be created at the beginning of each month. Given that the implied volatility of the index options is mostly between 30-50%, the option premium received each month would mostly be around 4% to 5%. Part of this premium will be paid out to the investor of the ETF as a regular dividend distribution.

When the index is going up, the upside of this performance cannot be recognized in the ETF because of the short call position. However, given the high option premium received per month (which would increase the net-asset-value of the ETF), it can compensate for small movements of the index during the month. The graph below shows the performance of this ETF since launch as of Sep 2025, (the levels are re-based, taking into account of the dividends received):



6.3.2. Adaptive Market Hypothesis

In a series of papers, Professor Andrew Lo (MIT) has developed a new investment paradigm which is known as the Adaptive Market Hypothesis. While the traditional investment framework aims to discover universally true investment strategies, analogous to physical laws, Lo suggests that no such framework can ever be complete. Efficient market hypothesis can only be correct under stable environments. However, when the environment is highly dynamic, as seen in the markets since the beginning of the 21st century, passive index investing cannot be the optimal strategy. Instead, strategies need to be dynamic in order to adapt to the changing market conditions.

Lo proposes to make use strategies that can manage risk via active volatility scaling algorithms. However, the strategy must be transparent and rule-based, so no discretionary investment judgment is needed. A simplified example is given in Lo(2016). In a strategy with a volatility controlled value-weighted index, a portion of the fund is invested in cash if the estimated volatility of the index at date t exceeds a pre-determined level, and a leveraged investment of more than 100% of the fund is made if the volatility falls below this level. In other words,

$$\widetilde{R_t} = \kappa_t R_t, \kappa_t = \min\left[\frac{\sigma_0}{\hat{\sigma}_{t-q}}, \bar{l}\right], \hat{\sigma}_{t-q}^2 = \frac{1}{n-1}\sum_{j=q}^{q+k-1}\left(R_{t-j} - \hat{\mu}_{t-q}\right)^2, q, \bar{l} \geq 1$$

σ_0 is a target volatility level, $\hat{\sigma}_{t-q}$ is the volatility estimator with a k-day rolling window, $\widetilde{R_t}$ is the index exposure, controlled by a leverage factor κ_t, $\hat{\mu}_{t-q}$ is the rolling-window mean return between $t-q-k+1$ and $t-q$, and $\bar{l}$ is some fixed upper bound on the amount of leverage that can be used.

Lo's paper applies this algorithm using CRSP value-weighted index returns from 1925 to 2014 with a 21-day rolling window volatility estimator with one lag, and a target volatility level of 16.9%. The maximum leverage is set to be 1.3. In the left diagram below, a 125-day rolling window is used, and the time series of returns is shown to be much less volatile than the unadjusted index. The green bar displays the leverage factor, and most of the time κ_t has been set to this maximum value, implying that the short-term annualized US equity market volatility is less than 16.9%/1.3 = 13%. The right diagram shows a comparison of the cumulative return over a 89-year period. The volatility-controlled index outperforms the raw index by a factor of 4.

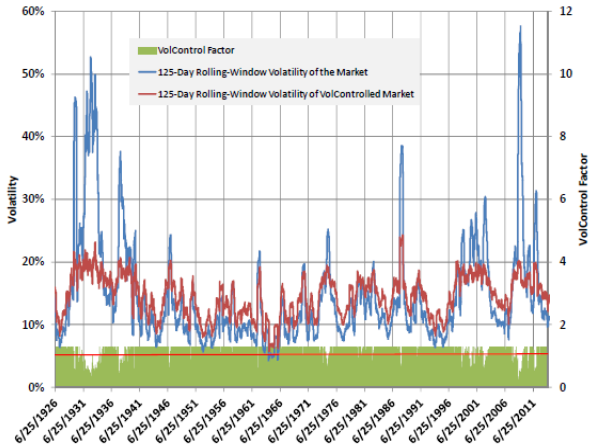


Figure 3. 125-day rolling-window annualized volatility estimates for CRSP daily value-weighted index returns from June 25, 1926 to December 31, 2014 with and without volatility control using a scale factor $\kappa_t = \text{Min}\left[\frac{16.9\%}{\hat{\sigma}_{t-1}}, 1.3\right]$, where $\hat{\sigma}_{t-1}$ is a 21-day rolling-window annualized volatility estimator.

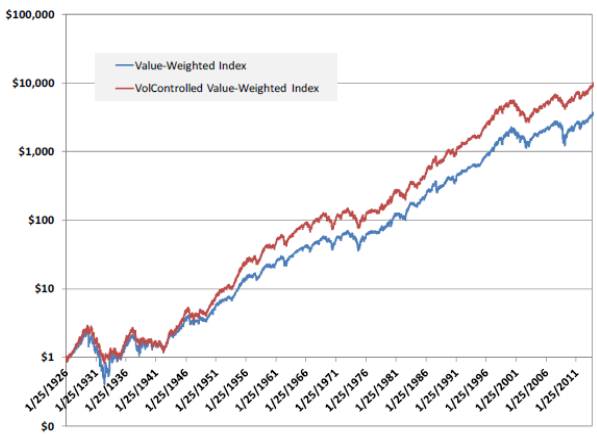


Figure 4. Cumulative return of the CRSP value-weighted index with and without volatility control from January 25, 1926 to December 31, 2014 (logarithmic scale). The volatility control consists of multiplying daily returns by a scale factor $\kappa_t = \text{Min}\left[\frac{16.9\%}{\hat{\sigma}_{t-1}}, 1.3\right]$, where $\hat{\sigma}_{t-1}$ is a 21-day rolling-window annualized volatility estimator of the value-weighted index return.

- Figures above taken from Lo (2016)

6.3.3. ESG investing

In recent years, a popular investment theme is so called ESG investing. ESG stands for "Environmental, Social and Governance". Asset managers and investors assess the ESG performance of different companies. Some advantages of each of these areas are:[7]

- Environmental factors
  - Reducing waste is economical.
  - Consumers will pay more for responsible products.
  - Reduces legal and other risks.
- Social factors
  - Good working conditions make employees more productive and attract talents.
  - Firms who don't discriminate attract more diverse talent.
  - Stocks with higher employee satisfaction perform better.
- Governance
  - Well governed firms perform better.
  - Companies with conservative accounting policies perform better.

However, it is arguable whether companies with good ESG performance may represent better investment choices, and there is a growing literature on the validity of this claim.

ESG investing is especially popular in Europe, but there has been substantial growth in other developed markets, including the US and Japan. This investment theme has also developed rapidly in China, although what kinds of companies can be qualified as satisfying ESG criteria are debatable.

6.3.3.1. ESG Index

Investments in ESG strategies are often made through ETFs and many benchmark indices have been created. In Hong Kong, Hang Seng Indexes Company Limited (the publisher of Hang Seng Index) introduced the first ESG index in 2010. An independent body, Hong Kong

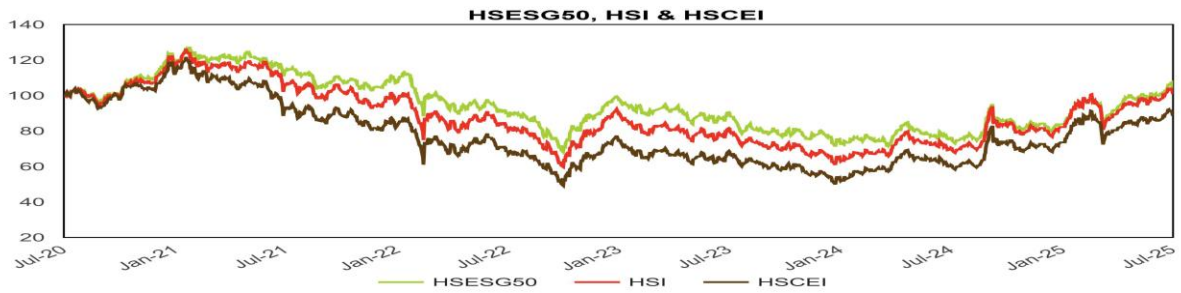
[7] L.H. Pedersen, "Responsible Investing: The ESG-Efficient Frontier", Presentation materials, AQR Capital Management, Aug 2021.

Quality Assurance Agency, evaluates approximately 500 Hong Kong listed companies and more than 1200 A-shares companies with a proprietary sustainability assessment and rating framework. Each company will receive an aggregate score representing its performance in seven areas: corporate governance, human rights, labour practices, environment issues, fair operating practices, consumer issues, community involvement and development. The score will be used to tilt the weighting of the company compared to the standard index.

As an example, we can compare the performance of the HSI ESG 50 and the Hang Seng Index. HSI ESG 50 includes 50 companies with the top ESG scores, choosing from stocks with relatively high market-capitalization. The following table shows the weights of the 10 biggest stocks in both HSI and HSI ESG 50 as of July 31, 2025:

	HSI		HSI ESG 50		
	Name	% weight	Name	% weight	Rank in HSI
1	Tencent	8.27	AIA	10.41	6
2	HSBC	7.93	Tencent	10.35	1
3	Alibaba	7.72	HSBC	9.91	2
4	Xiaomi	6.14	China Mobile	8.87	8
5	China Construction Bank	5.54	Hong Kong Exchanges	8.54	9
6	AIA	5.00	BYD	7.06	11
7	Meituan	4.26	SMIC	3.82	17
8	China Mobile	3.40	Techtronic Industries	2.30	25
9	Hong Kong Exchanges	3.27	CKH Holdings	2.29	23
10	ICBC	3.16	CLP Holdings	2.29	24

Two points are worth noting. Firstly, we notice that the ranks in the two indices can be different. For example, the weight of Tencent is higher than AIA in HSI, but it is lower in HSI ESG 50. This is an indication of a better ESG performance of AIA, resulting in extra weights. Secondly, many major companies are not included in the HSI ESG 50 (such as Alibaba, Xiaomi, Meituan, CCB). This may be a major cause of the different performance of the two indices.



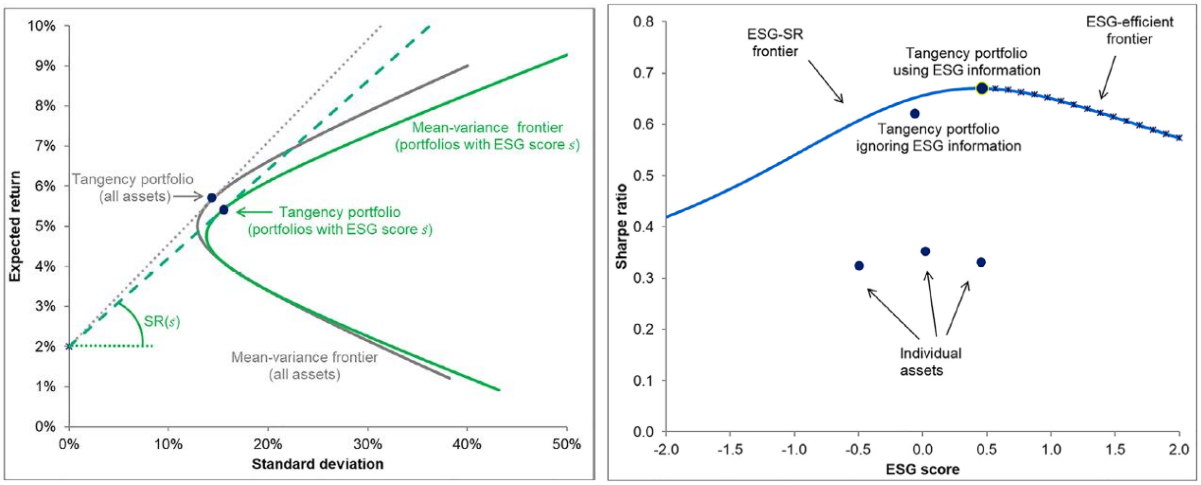
Index	Index Level	% Change						
		1 - Mth	3 - Mth	6 - Mth	1 - Yr	3 - Yr	5 - Yr	YTD
HSESG50	3,243.66	+3.89	+12.34	+24.04	+36.97	+15.47	+3.91	+23.45
HSI	24,773.33	+2.91	+12.00	+22.49	+42.83	+22.90	+0.72	+23.50
HSCEI	8,882.95	+2.36	+9.99	+20.33	+45.45	+29.01	-11.52	+21.85

6.3.3.2. ESG and investment performance

If we don't consider the social impact, investors would be most concerned whether investments in companies with good ESG scores can produce optimal results. By definition, if additional constraints are introduced in the optimization process, it would not be possible to achieve the performance of the unconstrained optimal portfolio as defined in Modern Portfolio Theory. However, it is still possible that including ESG information can raise expected returns while the increase in risk is only modest.

Pederson et al. (2021) propose a theoretical framework to understand the impact of ESG on Markowitz portfolio selection and CAPM. Given that the risk free asset is available and there are N risky assets with returns E(r), σ(r), covariance, and ESG scores, the market consists of three types of investors:

- ESG-unaware (Type-U): use E(r), σ (r) and covariance only; Markowitz optimization.
- ESG-aware (Type-A): Update E(r) and σ (r) with ESG information; Markowitz optimization.
- ESG-motivated (Type-M): use ESG information together with E(r), σ (r) and covariance to perform the optimization.

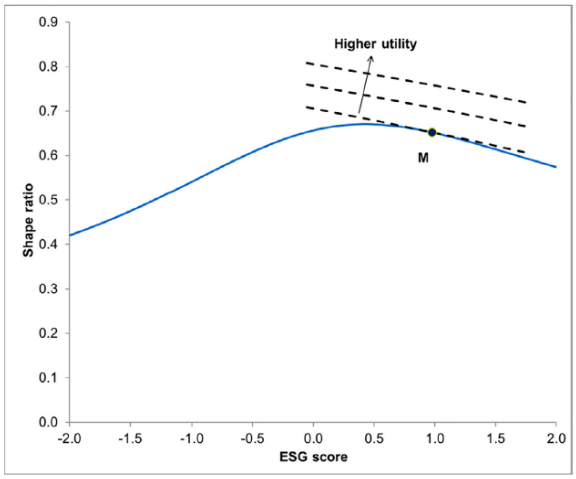


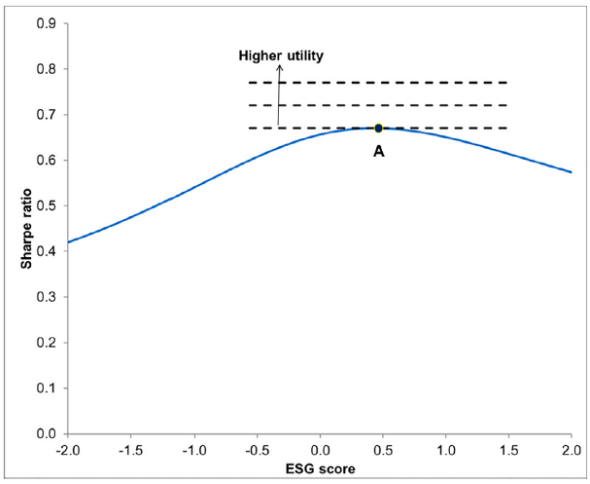
The black line on the left diagram shows the minimum variance frontier from a standard Markowitz optimization for a type-A investor. The tangency portfolio (all assets) is the portfolio with the highest Sharpe ratio. This portfolio corresponds to a particular ESG score (called this portfolio T). When constraints on ESG are added to the portfolio optimization exercise, the results would fall inside the feasible region of all possible portfolios. One result is shown in the green line on the left diagram. The tangency portfolio that can be achieved with this subset would have a lower Sharpe ratio.

The right diagram (known as the ESG-SR frontier) plots the Sharpe ratio with the ESG score, which shows a hump shape. This can be explained as follows. The absolute maximum Sharpe ratio corresponds to portfolio T. When ESG score varies, the Sharpe ratio will decrease, as given by the Sharpe ratios as obtained from the green lines. The points on the right of portfolio T are more efficient than the points on the left of portfolio T, given that a portfolio with a higher ESG score but with the same Sharpe ratio is preferable to a portfolio with a lower ESG score (therefore only the right portion of the curve constitutes

the ESG efficient frontier). If ESG information is ignored (e.g. for a type-U investor), the optimization exercise would be sub-optimal as less information is included. Hence the resulting tangency portfolio would have a maximum Sharpe ratio below the highest achievable Sharpe ratio for the type-A investor, shown in a point below the ESG-SR efficient frontier.

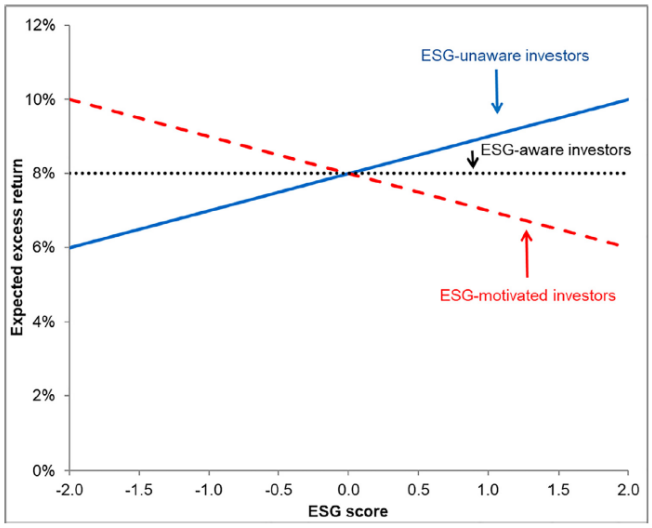
Investors could then choose the optimal portfolio by superimposing the indifference curves onto the ESG-SR frontier (c.f. section 4.3.4).





Clearly, the indifference curves for type-M investors are downward sloping, as they would prefer portfolios with higher ESG scores in lieu of higher Sharpe ratios. Type-A investors would be indifferent to the actual ESG scores as the information should already be incorporated into the estimates of expected returns and variances. For these investors, the higher the achievable Sharpe ratio the better, and ESG scores do not matter. Hence the indifference curves are horizontal.

Finally, Pederson et al. (2021) also propose an ESG-adjusted CAPM.



The shape of the curve depends on which type of investor dominates the market. When there are many type-U investors and higher ESG scores predicts higher future profits for the companies, there is a premium for ESG investments as the prices of these companies have not been bid up, and hence the expected future returns are higher such that the (blue)

line is upward sloping. When the economy has many type-A investors, these investors would already bid up the prices of high-ESG socks to reflect their expected profits, and thus ESG scores would no longer be connected to expected excess return, giving a horizontal profile (as in the blacked doted line). Finally, if the economy has many type-M investors, higher ESG-stocks actually deliver lower expected excess returns, but these investors are willing to accept a lower return for a higher ESG portfolio. Thus, the profile (red line) is downward sloping.

This theoretical framework can be tested with empirical data, but much work remain.

Other References

Doyle, J. and G. Hayman. *Smart Beta, Direct Indexing, and Index-Based Investment Strategies: A Framework*. Research and Policy Center, CFA Institute, July 2024.

Lo, A. "What is an Index?" *Journal of Portfolio Management* 42(2):21-36, 2016.

Lo A. & MacKinlay A.C. *A Non-random walk down Wall Street*. Princeton, 2001.

Pedersen, L.J., Fitzgibbons, S. and Pomorski, L., "Responsible Investing: The ESG-efficient Frontier," *Journal of Financial Economics* 142 (2021):572-597.